



CSIS/CSIQ1624 (2025)
Semester Test 1 (Theory & Practical)

Assessor(s): Mr J. Viera
 Dr M.B. Mase

Date: 1 September 2025

Moderator: Prof T. Stott

Time: 2h30min (15m for submission)

Marks: 70

Instructions

- All files must include your name and student number at the very top of every file. Marks will be subtracted for every missing occurrence (2 marks).
- The computer monitor must be switched off while you are completing Section A and B. Answer Section A and B on the answer sheet that is provided.
- This is a closed-book test.
- This semester test consists of three sections: Theory (Section A), Code comprehension (Section B) and Visual Studio project (Section C).

SECTION A (20)

1. In each case, state whether the following statement is **true** or **false**.
 - 1.1 Objects do not belong to classes. (1)
 - 1.2 The keyword **this**, refers to the current state instance of a class Form. (1)
 - 1.3 Objects have properties that describe their appearance and behaviour. (1)
 - 1.4 During the initialization of jagged arrays, the first square bracket specifies the size of an array. (1)
 - 1.5 Generic lists are not defined with the type or class of its members between **<>**. (1)
 - 1.6 Arrays are useful when you need to store just one value in memory at a time. (1)
 - 1.7 The single comma [,] represents the array as two-dimensional. (1)
 - 1.8 A jagged array is one-dimensional array. (1)
 - 1.9 StreamReader class can be used to read data from files on disk. (1)
 - 1.10 The elements of a jagged array cannot be of different dimensions and sizes. (1)

2. Answer the following multiple-choice questions by choosing the correct answer.
- 2.1 What is the control that provides information or instructions? (1)
- A. Label
 - B. Textbox
 - C. SaveFileDialog
 - D. OpenFileDialog
- 2.2 Which control allows user to enter character string during the run-time? (1)
- A. PictureBox
 - B. Button
 - C. TextBox
 - D. Label
- 2.3 Which piece of code is executed when an event occurs, for example when the user clicking the button? (1)
- A. Event
 - B. MessageBox
 - C. Object
 - D. Event handler
- 2.4 Which of the following generic method remove the element at the specified index? (1)
- A. Clear()
 - B. Remove()
 - C. RemoveAt()
 - D. Reverse()
- 2.5 Which property gets or sets the number of elements that the array list actually contains? (1)
- A. Text
 - B. Capacity
 - C. BackColor
 - D. Count
- 2.6 What is the purpose of the StreamWriter class? (1)
- A. Write text data to a file
 - B. Read text data from a file
 - C. Return an objects
 - D. Directory of source files
- 2.7 What is another way to ensure that a file is closed automatically when an object goes out of the scope of StreamWriter/StreamReader object? (1)
- A. **if...else** statement
 - B. **while** loop
 - C. **using** statement
 - D. Event statement

- 2.8 Which namespace contains classes that enable users to write and read from files on disk? (1)
- A. using System.Windows.Forms
 - B. using System.Drawing
 - C. using System.Data
 - D. using System.IO
- 2.9 What are jagged arrays? (1)
- A. A jagged array is an array of arrays.
 - B. An array that can be indexed as though it were a single line.
 - C. Are arrays indexed by two subscripts, one for the row and one for the column.
 - D. Are arrays without square brackets.
- 2.10 How do you declare a jagged array of integers? (1)
- A. int [] Array
 - B. int [][] Array
 - C. int [,] Array
 - D. int [, ,] Array

[20]**SECTION B (10)**

The following two questions focus on code comprehension. You **may not** use Visual Studio for this section of the test.

Question 1 (5)

Consider the following code snippet and answer the questions that follow:

```
1.  int[] Numbers = new int[] { 1, 2, 3, 4, 5, 6 };
2.  int iSum = 0;
3.  Array.Reverse(Numbers);
4.  for (int i = 0; i < Numbers.Length; i++)
5.  {
6.      iSum += Numbers[i];
7.      Console.Write(Numbers[i] + " ");
8.  }
9.  Console.WriteLine("\nThe sum is: " + iSum);
10. Console.ReadLine();
```

1. What data type is the one-dimensional array? (1)
- A. string
 - B. int
 - C. decimal
 - D. double
2. What is the length of the one-dimensional array? (1)
- A. 5
 - B. 4
 - C. 7
 - D. 6

3. What is the purpose of the reverse method? (1)
- A. Sort the elements or a portion of them
 - B. Reverse the order of the elements
 - C. Sets the number of elements that the array list actually contains
 - D. Returns the index of the first occurrence of a value
4. What is the sum of the array list? (1)
- A. 20
 - B. 19
 - C. 15
 - D. 21
5. How are the elements going to be display when you run your application? (1)
- A. 1 2 3 4 5 6
 - B. 6 5 4 3 2 1
 - C. 1 3 2 6 5 4
 - D. 1 6 5 4 3 2

Question 2 (5)

Consider the following code snippet and answer the questions that follow.

```
1. int[,] numbers = { { 1, 2, 3, 0 }, { 4, 5, 6, 8 }, { 2, 8, 9, 7 } };
2. for (int r = 0; r < numbers.GetLength(0); r++)
3. {
4.     Console.Write("Row" + r + " ");
5.     for (int c = 0; c < numbers.GetLength(1); c++)
6.     {
7.         Console.Write(numbers[r, c] + " ");
8.     }
9.     Console.WriteLine();
10. }
11. Console.ReadLine();
```

1. How many rows and columns does the given two-dimensional array has? (1)
- A. 3 rows and 4 columns
 - B. 4 rows and 3 columns
 - C. 2 rows and 3 columns
 - D. 5 rows and 4 columns
2. What the total length of the given elements? (1)
- A. 15
 - B. 9
 - C. 10
 - D. 12
3. Which indexes to use to access 5? (1)
- A. numbers [1,1]
 - B. numbers [1,0]
 - C. numbers [0,3]
 - D. numbers [0,1]

4. Which indexes to use to access 7? (1)
- A. numbers [1,1]
 - B. numbers [1,0]
 - C. numbers [0,3]
 - D. numbers [2,3]
5. Which of the following will be the output of the given two-dimensional array? (1)
- A.

1	2	3	0
4	5	6	8
2	8	9	7

B.

1	2	3	0
2	8	9	7
4	5	6	8
- C.

1	2	3	0
4	0	0	8
2	8	1	7

D.

2	8	9	7
4	5	6	8
1	2	3	0

SECTION C (40)
Instructions:
1. Save to the T Drive:

- Create a new folder within the "T" drive to hold your C# projects. Name the folder appropriately (**SudentNumber_SemesterTest1**). Save all your projects in this folder. ZIP your folder and submit through BlackBoard.
- We are blind to green. We don't mark the code that is commented out.
- It is your own responsibility to save your work often. NO extra time will be given if you were to lose your work!

2. Adhere to Conventions:

- Follow the programming and naming conventions you have been taught throughout your course.

3. Proper Alignment and Formatting:

- Review the code and ensure that it is properly aligned and formatted. Do not add excessive white space, but make sure the code is readable.

Question 1 (20)

Develop console application (C#) using Visual Studio 2022 and name it **Question 1**. The application should print a list of students who performed well in programming module from first to third year.

```
Best performers in Computer Science:

Year 1 students:
*****
Ben
Merry
Dean

Year 2 students:
*****
Pulane
Peter
Marriam
Daniel

Year 3 students:
*****
Maggie
Gert
```

Figure 1: List of students

- 1.1 Declare and instantiate the jagged array. The name of your jagged array should be **students**. (3)
 - 1.2 Initialize all the elements of your jagged array upon declaration. (6)
 - 1.3 Display all the names of all the students according to their level of education (see **Figure 1**). (11)
- [20]**

Question 2 (20)

Develop Windows Form application (C#) using Visual Studio 2022 and name it **Question2**. The application should allow the user to **Save** and **Read** any student questions. The student questions should be saved in the "**Questions.txt**" file in the **Debug** subfolder. The screen prints in **Figure 2** and **Figure 3** illustrates what is expected.

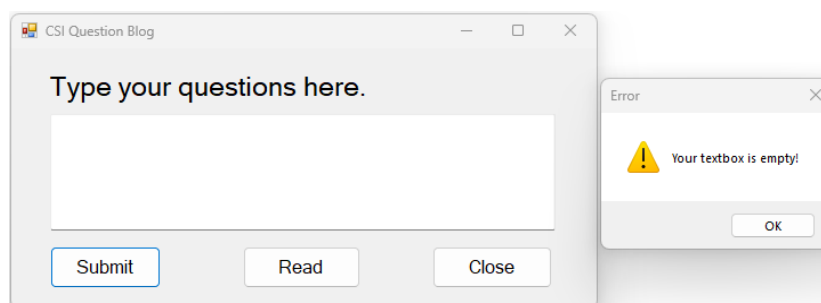
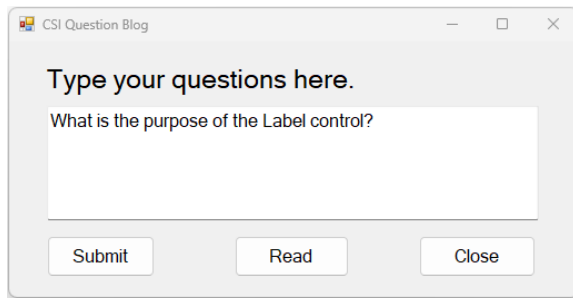


Figure 2: Error message

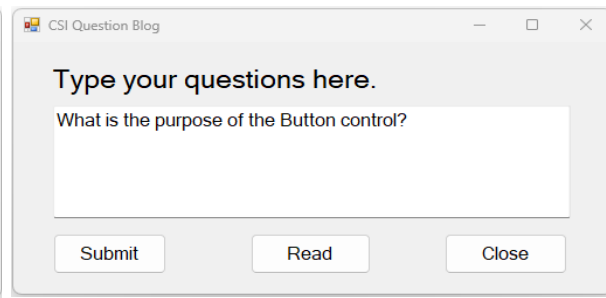


CSI Question Blog

Type your questions here.

What is the purpose of the Label control?

Submit Read Close

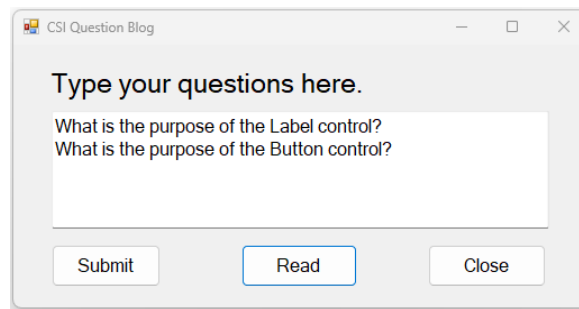
Figure 3: First question


CSI Question Blog

Type your questions here.

What is the purpose of the Button control?

Submit Read Close

Figure 4: Second question


CSI Question Blog

Type your questions here.

What is the purpose of the Label control?
What is the purpose of the Button control?

Submit Read Close

Figure 5: Two questions displayed

Your controls should be named according to the following:

- **Buttons:** btnSave, btnRead & btnClose
- **Label:** lblQuestion
- **Textbox:** txtQuestion
- **Form:** frmCSIQuestionBlog

Develop your application according to the instructions that follow.

- 2.1 Form creation and all controls added. (2)
- 2.2 When the user selects the **Submit** button, the student question must be saved to a text file (named **Questions.txt**). The content of the file must not be overwritten. Clear the textbox when the student question is successfully saved. **Note:** The **Questions.txt** must be saved in the **Debug** folder of your project. (9)
- 2.3 When the user selects the **Read** button, the student question/s should be displayed in the text box (see **Figure 5**). (4)
- 2.4 When the user selects the **Close** button, the form should be closed. (1)
- 2.5 Use defensive programming to avoid run-time errors. (4)

[20]

Total Marks = 70

END OF SEMESTER TEST 1

Main Test 1 - Memo

SECTION A (20)

Question 1

1. In each case, state whether the following statement is **true** or **false**.

- 1.1 False ✓
- 1.2 True ✓
- 1.3 True ✓
- 1.4 True ✓
- 1.5 False ✓
- 1.6 False ✓
- 1.7 True ✓
- 1.8 False ✓
- 1.9 True ✓
- 1.10 False ✓

2. Answers for multiple-choice questions.

- 2.1 A ✓
- 2.2 C ✓
- 2.3 D ✓
- 2.4 C ✓
- 2.5 D ✓
- 2.6 A ✓
- 2.7 C ✓
- 2.8 D ✓
- 2.9 A ✓
- 2.10 B ✓

SECTION B (10)

The following two questions focus on code comprehension.

Question 1 (5)

1. B ✓
2. D ✓
3. B ✓
4. D ✓
5. B ✓

Question 2 (5)

1. A ✓
2. D ✓
3. A ✓
4. D ✓
5. A ✓

SECTION C (40)

Question 1 (20)

Jagged Array Application

```
static void Main(string[] args)
{
    string[][] students = new string[3][]
    {
        new string[] { "Ben", "Henry", "Dean" },
        new string[] { "Pulane", "Peter", "Max", "Marian", "Daniel" },
        new string[] { "Maggie", "Gert" }
    };
    Console.WriteLine("Best performers in Computer Science:" + "\n");
    for (int i = 0; i < students.Length; i++)
    {
        Console.WriteLine("Year {0} students: ", i + 1);
        Console.WriteLine("*****");
        for (int j = 0; j < students[i].Length; j++)
        {
            Console.Write(students[i][j] + "\t" + "\n");
        }
        Console.WriteLine();
    }
    Console.ReadLine();
}
```

(20)

Question 2 (20)

CSI Question Blog application

Form ✓✓

```

using System;
using System.IO;
using System.Windows.Forms;

namespace Question2
{
    public partial class frmCSIQuestionBlog : Form
    {
        public frmCSIQuestionBlog()
        {
            InitializeComponent();
        }
        private void btnClose_Click(object sender, EventArgs e)
        {
            this.Close();
        }
        private void btnSubmit_Click(object sender, EventArgs e)
        {
            try
            {
                if (txtQuestion.Text != "")
                {
                    StreamWriter f = new StreamWriter("Questions.txt", true);
                    f.WriteLine(txtQuestion.Text);
                    f.Close();
                    txtQuestion.Clear();
                    txtQuestion.Focus();
                }
                else
                {
                    MessageBox.Show("Your textbox is empty!", "Error",
                        MessageBoxButtons.OK, MessageBoxIcon.Warning);
                }
            }
            catch (Exception ex)
            {
                MessageBox.Show(ex.Message);
            }
        }

        private void btnRead_Click(object sender, EventArgs e)
        {
            try
            {
                txtQuestion.Clear();
                StreamReader f = new StreamReader("Questions.txt");
                txtQuestion.Text += f.ReadToEnd();
                f.Close();
            }
            catch (Exception ex)
            {
                MessageBox.Show(ex.Message);
            }
        }
    }
}

```

(20)

